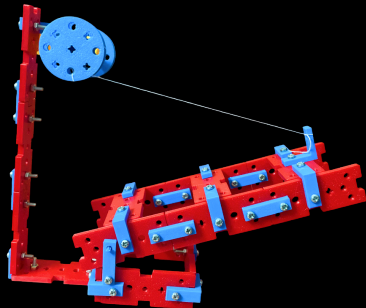


level 2

no. of sessions : 5

drawbridge

The sensor-controlled drawbridge opens when a boat approaches and closes after it passes, just like bridges over rivers that let ships cross.

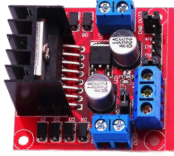


simple machine: pulley

electronics: input vs output

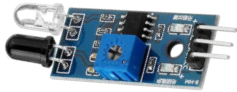
oh.





Motor Driver

The **brain of a robot**. I listen to input devices like sensors and tell output devices like motors/LEDs when to turn ON/OFF.



IR Sensor

I am like a **robot's eyes**. I help robots see objects or follow a path without touching anything.



LDR Sensor

I am like a **light detector** who tells a robot if it is bright or dark and i am used in automatic lights.



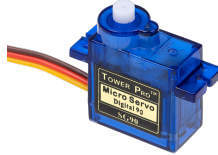
Motor

I **change electricity into movement** and help wheels, fans, and robots spin and move.



DPDT Switch Controller

Like a **direction button** - i make a motor spin forward or backward without using coding.



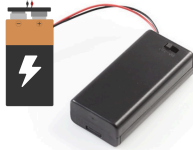
Servo Motor

I am a **smart motor** that can stop at the exact position you want. I am used in robot arms, gates, and moving parts.



Servo Controller

I am like a **control room** for the servo motor and lets you move the servo easily to test how it works.



Battery & Battery Holder

I **give power** to the project, and my battery holder keeps the battery safe and connected. Together, we help the model work.



Jumper Wires

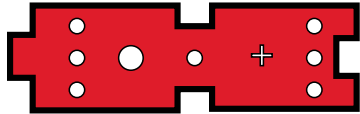
We are the **roads for electricity** who connect the different parts of the circuit, so power can travel.



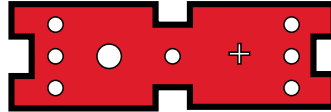
Motor Clamp

I am like a **strong hand** that holds the motor tightly in place while it spins.

materials needed



M-F beam - x4



F-F beam - x2



I clamp - x4

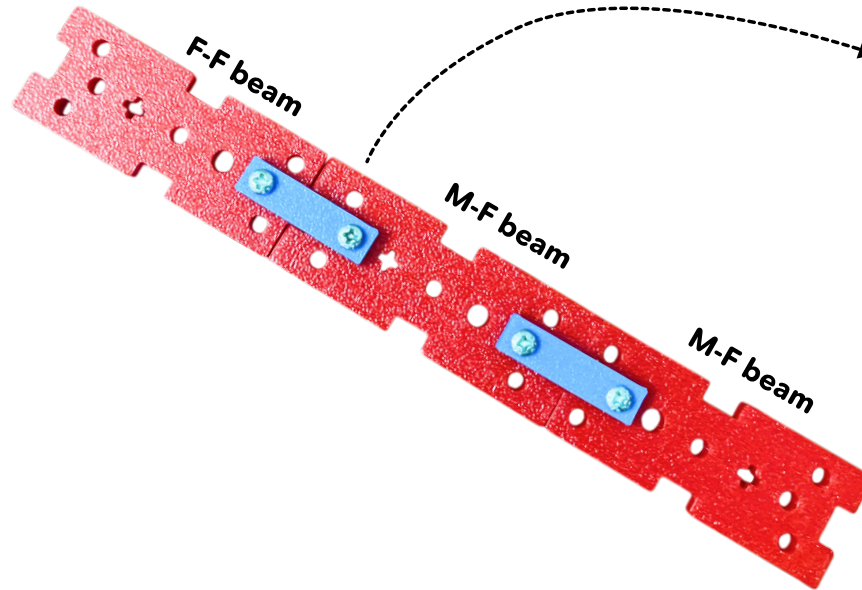


small screw - x8



nut - x8

step 1: assemble the body



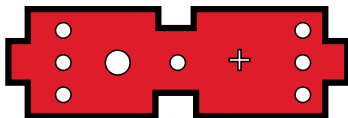
Lets join
2 x M-F beams
and a F-F beam.

Use 2 I-clamps to
hold them together.

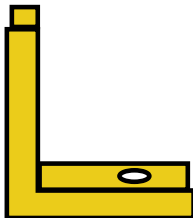


Repeat step 1 to make another set.

materials needed



M-M beam - x3



L clamp - x6

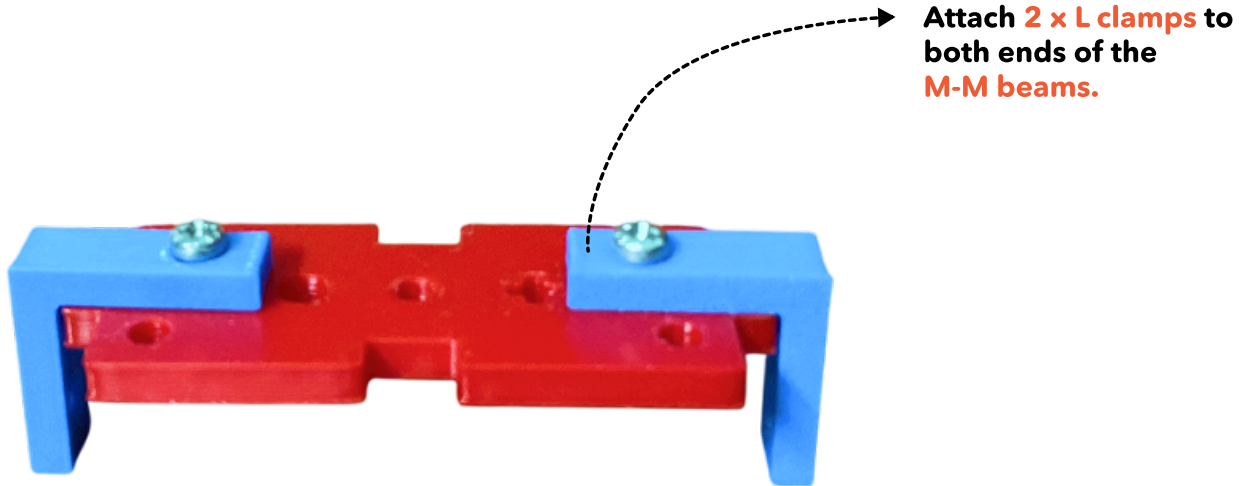


small screw - x6



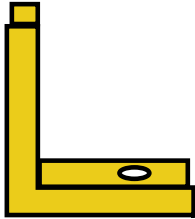
nut - x6

step 2: assemble the supports



Repeat this step to make 2 more sets.

materials needed



L clamp - x1



small screw - x7

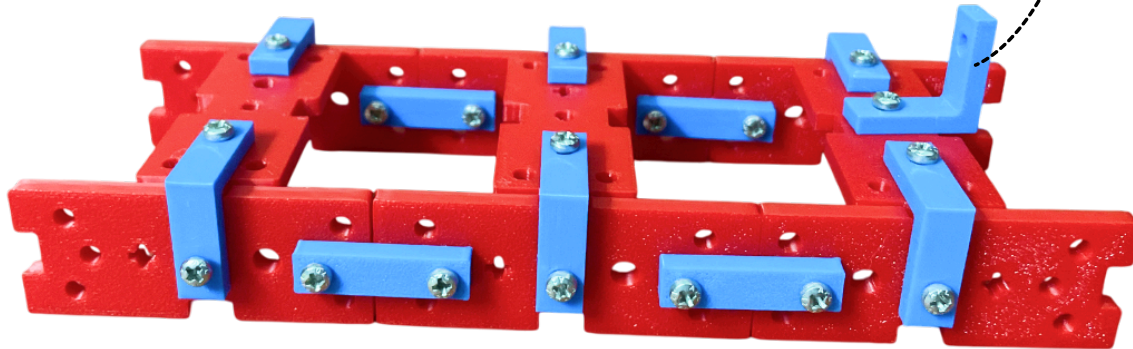


nut - x7

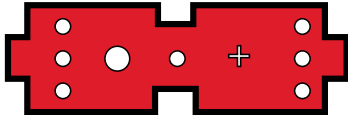
step 3: complete the bridge assembly

Join the pieces from
step 1 and 2 as shown.

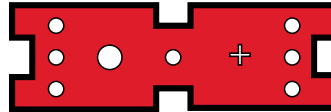
Attach a **L clamp**
to the support
as shown



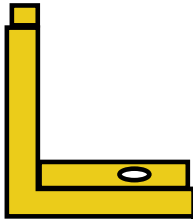
materials needed



M-M beam - x2



F-F beam - x2



L clamp - x5



small screw - x9

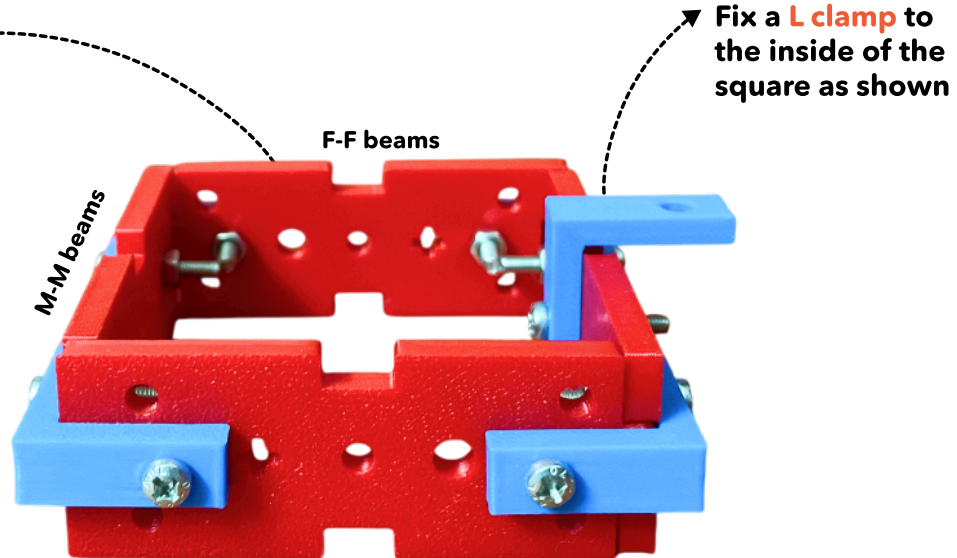


nut - x9

step 4: assemble the drawbridge's holder

Join **2 x F-F Beams** and **2 x M-M Beams** together to form a square.

Use **4 x L Clamps** to hold them together.



materials needed



I clamp - x2

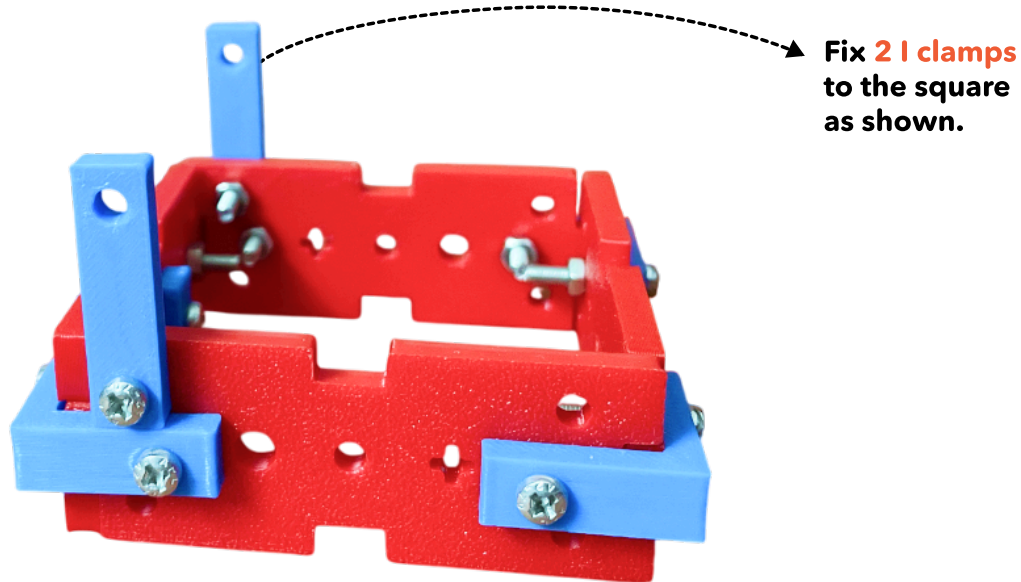


small screw - x2



nut - x2

step 5: assemble the axle holder

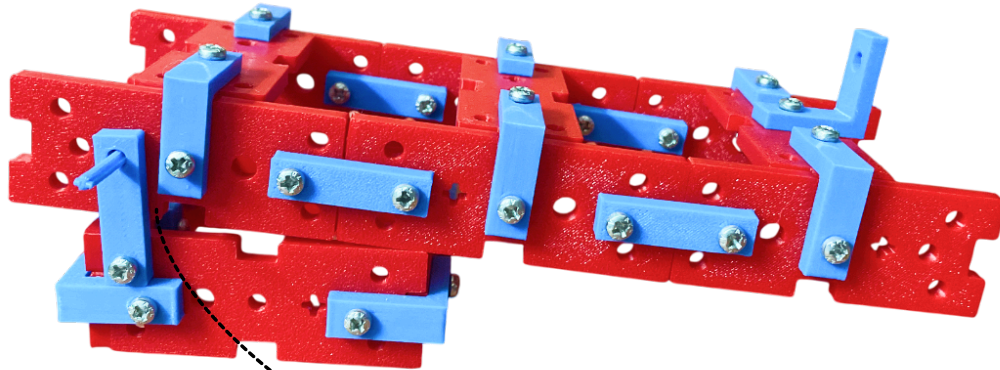


materials needed

long axle x1



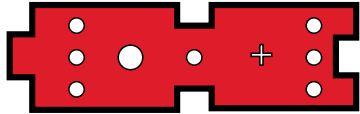
step 6: join the holder and bridge



→ Place the bridge in
between the holder.

Align the holes on the
I clamp and the bridge and
pass a **long axle** through it.

materials needed



M-F beam - x4



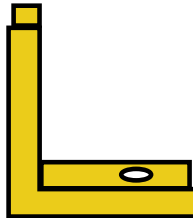
small screw - x7



nut - x7



I clamp - x2



L clamp - x2

step 7: assemble the motor supports

1.

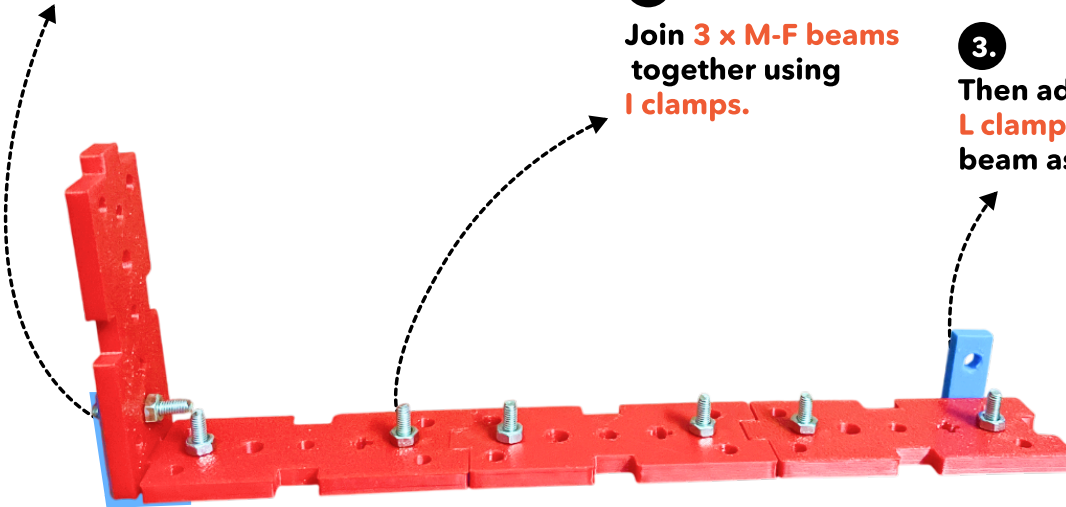
Attach a
M-F beam using
an **L clamp**

2.

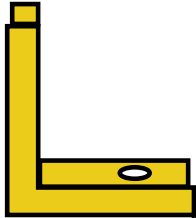
Join **3 x M-F beams**
together using
I clamps.

3.

Then add another
L clamp to the other
beam as shown.



materials needed



L clamp - x1

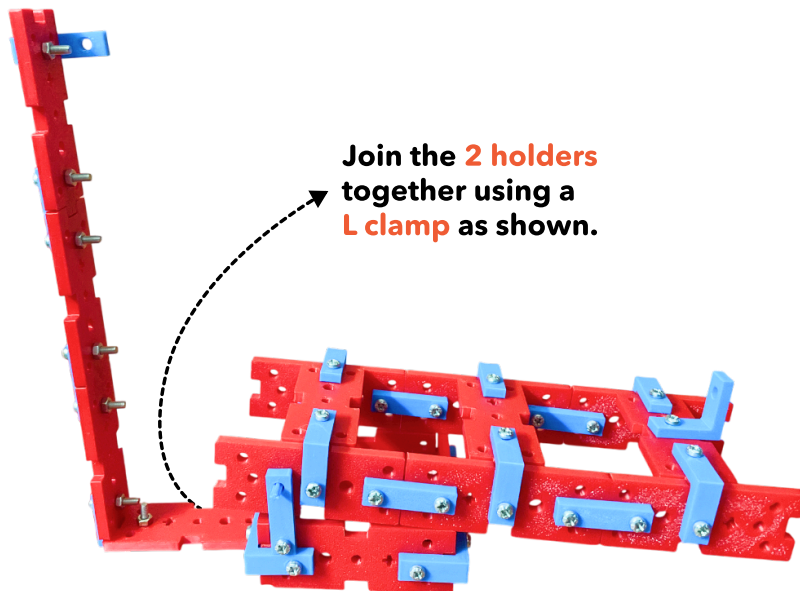


small screw - x2



nut - x2

step 8: attach the motor supports



materials needed



motor x1

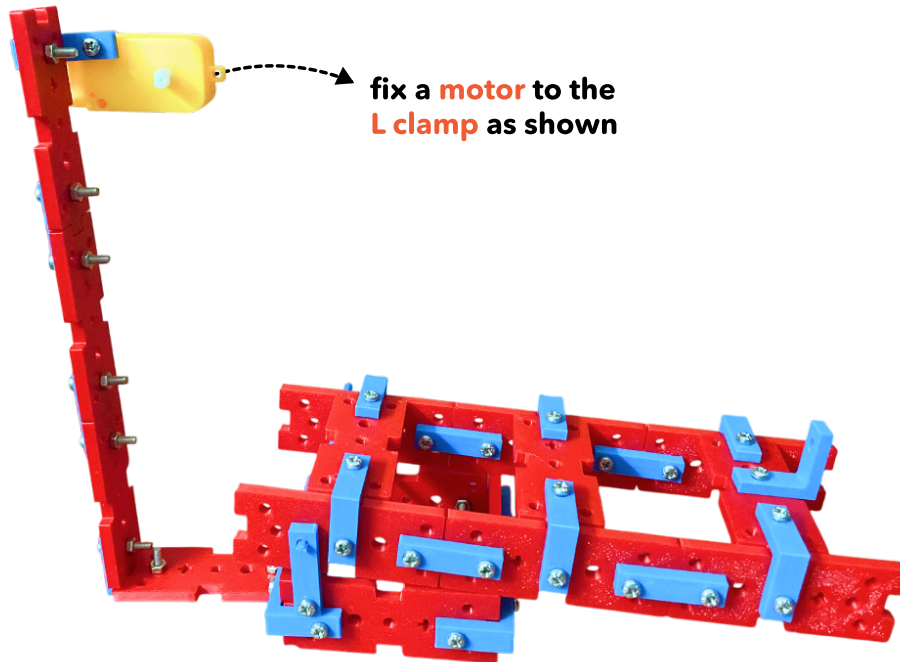


long screw - x1



nut - x1

step 9: add the motor



materials needed

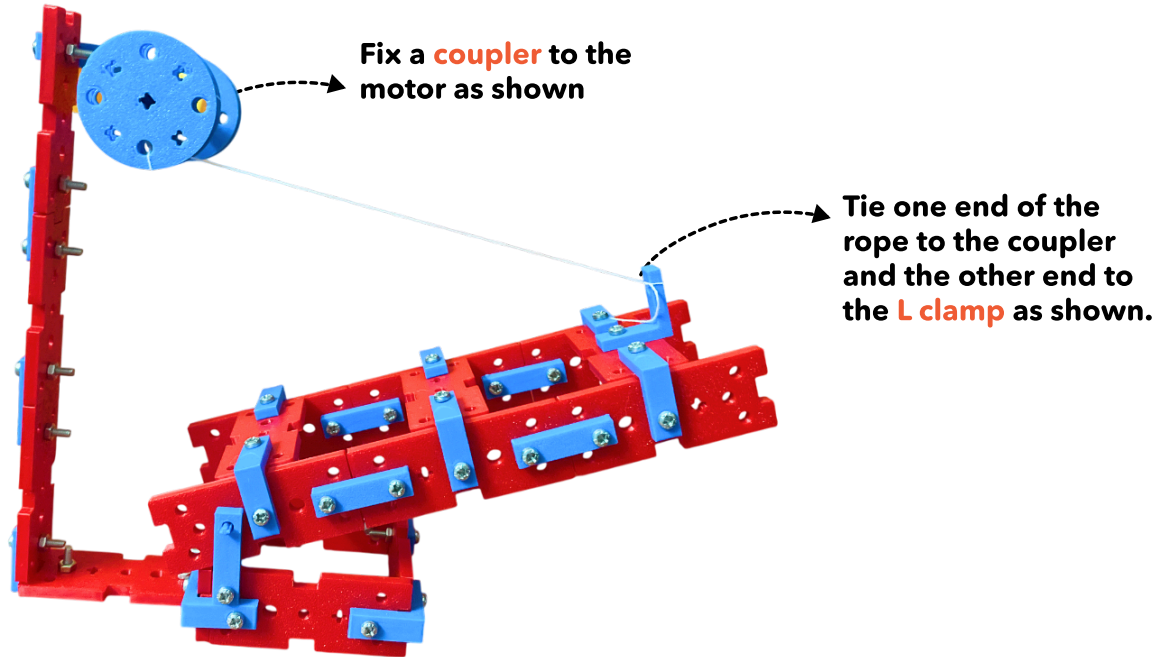


coupler - x1



rope x1

step 10: complete the pulley system



materials needed



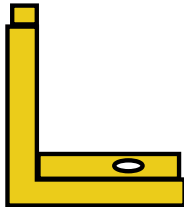
IR sensor x2



small screw - x2



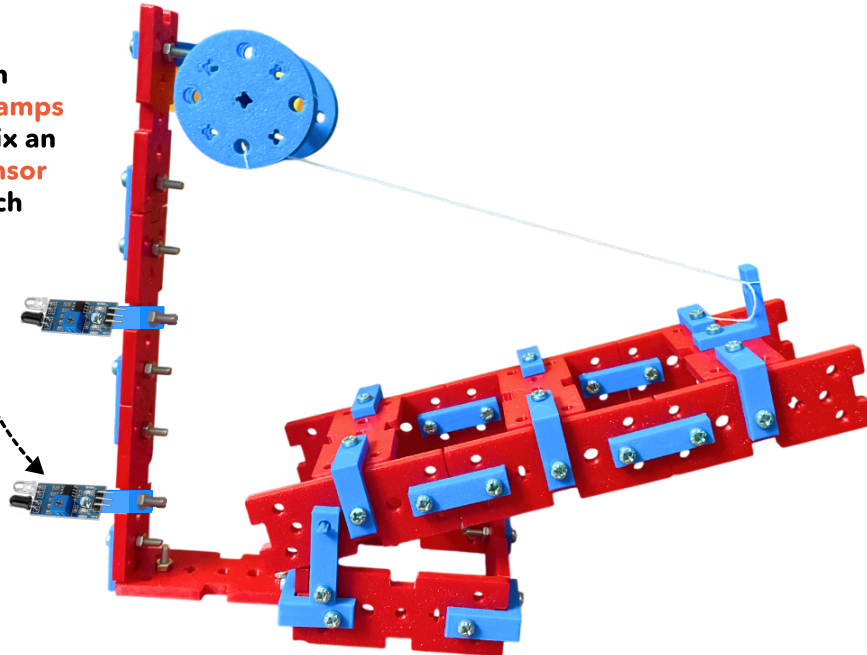
nut - x2



L clamp - x2

step 11: add the IR sensor

attach
2 L clamps
and fix an
IR sensor
to each



materials needed



IR sensor x2



small screw - x2



nut - x2



battery x1



battery holder x1



jumper cables

step 12: circuit assemble and testing

